Trần Văn Tài

AI Engineer

0383987056 vtai2805@gmail.com https://github.com/codecuatai Dong Da, Ha Noi

JOB OBJECTIVES

A highly motivated and detail-oriented developer seeking an AI Engineer Internship. Eager to apply my strong foundation in Python, backend development, and machine learning to build intelligent solutions while further developing my skills in real-world AI applications and system operations.

ACADEMIC

Thuy Loi University 2023 - 2027

Major: Information System

GPA: 3.77

WORK EXPERIENCE

EMS Fitness & Yoga 3/2026 - 4/2026

Role: AI Engineer

Developed an AI-powered chatbot integrated with Facebook Messenger for a gym using a Retrieval-Augmented Generation (RAG) architecture to provide customer consultation and answer inquiries, collect lead information, and automatically send real-time notifications to Telegram whenever a new lead was generated, enabling the sales team to promptly engage and follow up with potential customers.

FPT Information System (FPT IS) 4/2026 - 7/2026

Role: Platform Engineer

AI-Powered Observability & Self-Healing Platform

- Developed an AI-powered observability platform using Python (FastAPI, PydanticAI) and Gemini API to automate anomaly detection, root cause analysis (RCA), and incident remediation.
- Built a multi-agent RCA workflow by integrating KeepHQ, Alertmanager, Qdrant, and Kubernetes telemetry to identify root causes and recommend resolutions.
- Designed a distributed execution engine using Spring Boot, Go, and gRPC to execute automated runbooks with human approval and role-based permissions.
- Reduced unnecessary LLM requests by 80% through intelligent alert filtering while ensuring auditability and multi-tenant isolation.

SKILLS

Programming Languages	Python, HTML/CSS, JavaScript, PHP
Backend	Django, FastAPI
Frontend	React, Next.js
Databases	SQLServer, PostgreSQL, Redis, Qdrant
Other	Git, GitHub, GitLab, Docker, CI/CD

PROJECT

Learning Hub AI 3/2026 - Nay

Personal Project

Description: Designed and optimized a state-of-the-art Multimodal Retrieval-Augmented Generation (RAG) system utilizing multi-agent orchestrations, hybrid vector searches, and self-reflection loops to ensure accurate, context-grounded AI tutoring.

Key Contributions & Achievements:

- Built a stateful multi-agent workflow in LangGraph combining Intent Classification, Context Retrieval, Relevance Grading, Answer Generation, and Quality Evaluation.
- Designed a self-reflection evaluation loop where Gemini 1.5 Pro acts as an evaluator to check generated answers for hallucinations, triggering automatic retry & generation iterations if answers lack source grounding.
- Utilized hybrid search retrieval within Qdrant to search document chunks by mixing sparse and dense vector representations, yielding high relevance scores.
- Designed complex study agents including Quiz Generator (creating interactive multiple-choice questions) and Essay Grader (comparing essays against source texts to output structured JSON critiques).
- Integrated LangSmith for full-lifecycle observability, tracking trace latency, debugging agent routing nodes, and monitoring token utilization.

Link Demo: https://www.learninghubs.tech

Source: https://github.com/MULTIMODAL-AI-LEARNING-HUB/learning-hub-web

TLU Student Handbook Chatbot 12/2025 - 5/2026

Scientific Research - Thuy Loi University

Designed and implemented an AI chatbot system for querying school administrative regulations based on Retrieval-Augmented Generation (RAG) architecture to assist students in navigating complex document systems.

- Data Processing: Collected, cleaned, and chunked 12+ regulation documents (PDF, DOCX); constructed metadata for filtering by academic year.
- Vector Database: Applied Sentence-Transformers to generate embeddings and stored them in Qdrant; integrated Hybrid Search (combining Vector Search with BM25) to maximize retrieval recall and precision.
- Result Optimization: Developed a Reranking module using Cross-Encoder (ms-marco-MiniLM) to reorder and select the most relevant documents before feeding them into the LLM.
- AI Pipeline Development: Designed a complex processing flow incorporating Question Analysis, automatic Query Expansion, and Prompt Engineering to guide the LLM in generating accurate and well-structured responses.
- Technologies Used: Python, LangChain, FastAPI, Groq API (Llama 3), Qdrant, HuggingFace Transformers.
- Frontend & Integration: Developed a demo interface using HTML/JS and tested integration with a Laravel web application to display chat results.

Link Demo: https://student-chat-67ab.onrender.com/

Source: https://huggingface.co/Chatbot-TLU

Source: https://github.com/codecuatai/student_chat

Application of Machine Learning Models for Counterfeit Currency Detection 10/2025 - 12/2025

Personal Project

Built a machine learning pipeline from scratch to classify genuine vs. counterfeit banknotes using statistical image features. Implemented, tuned, and compared five algorithms (Logistic Regression, SVM, Decision Trees - ID3/CART, Perceptron) with custom gradient descent and regularization, achieving up to 98.9% accuracy. Conducted comprehensive data preprocessing, exploratory analysis, hyperparameter optimization, and cross-model evaluation to identify the best-performing approach.

Source: https://github.com/codecuatai/BTL_Banknote_Authentication_Dataset

ACTIVITIES

Engaged in algorithmic programming competitions at Thuy Loi University 2024

FAVOURITE

Running, Coding